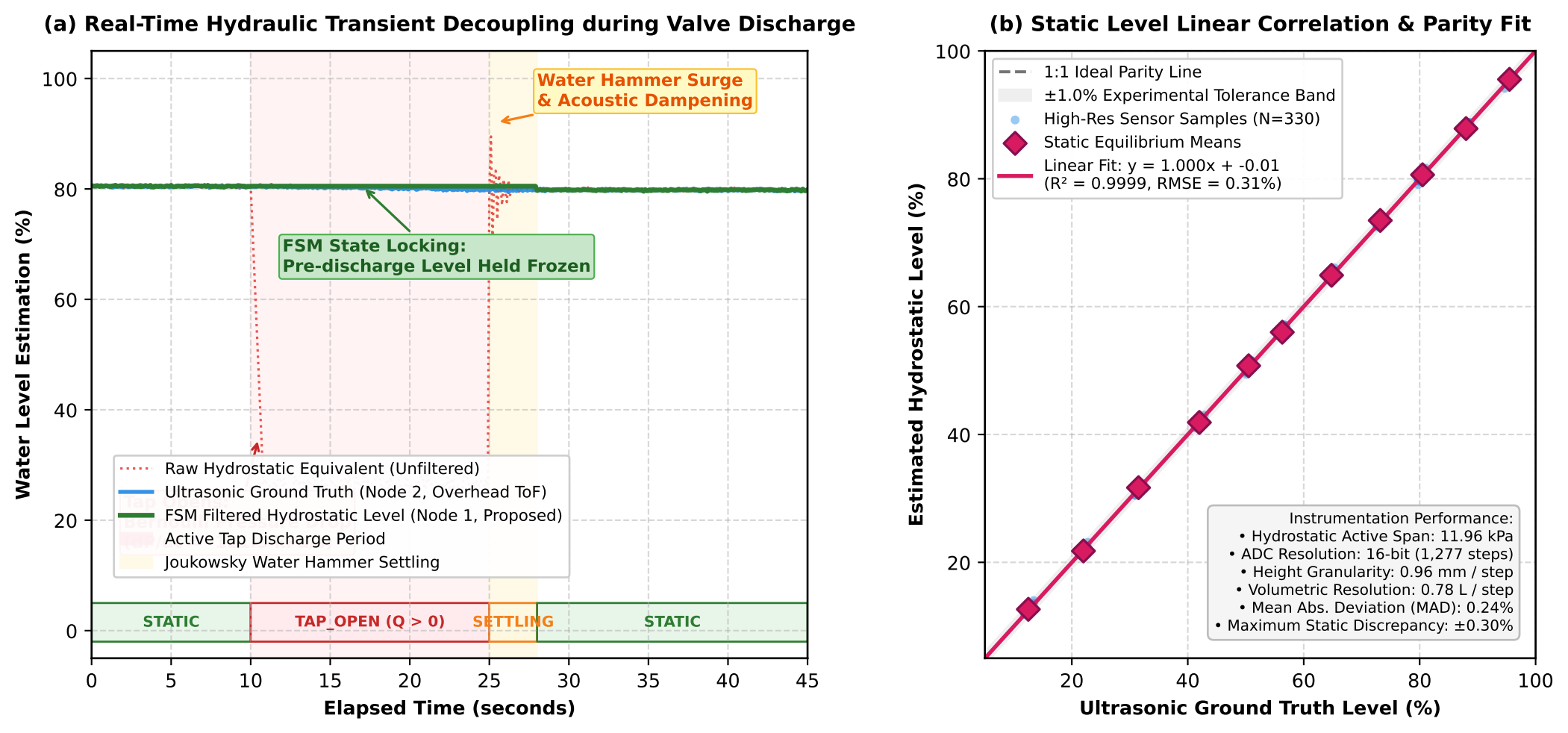

(a) Real-Time Hydraulic Transient Decoupling during Valve Discharge
Water Hammer Surge
& Acoustic Dampening
FSM State Locking:
Pre-discharge Level Held Frozen
Raw Hydrostatic Equivalent (Unfiltered)
Ultrasonic Ground Truth (Node 2, Overhead ToF)
FSM Filtered Hydrostatic Level (Node 1, Proposed)
Active Tap Discharge Period
Joukowsky Water Hammer Settling
STATIC
TAP_OPEN (Q > 0)
SETTLING
STATIC
Water Level Estimation (%)
Elapsed Time (seconds)
0
5
10
15
20
25
30
35
40
45
0
20
40
60
80
100
(b) Static Level Linear Correlation & Parity Fit
1:1 Ideal Parity Line
±1.0% Experimental Tolerance Band
High-Res Sensor Samples (N=330)
Static Equilibrium Means
Linear Fit: y = 1.000x + -0.01
(R² = 0.9999, RMSE = 0.31%)
Instrumentation Performance:
• Hydrostatic Active Span: 11.96 kPa
• ADC Resolution: 16-bit (1,277 steps)
• Height Granularity: 0.96 mm / step
• Volumetric Resolution: 0.78 L / step
• Mean Abs. Deviation (MAD): 0.24%
• Maximum Static Discrepancy: ±0.30%
Estimated Hydrostatic Level (%)
Ultrasonic Ground Truth Level (%)
20
40
60
80
100